

# RGIPT

Subject: rand

Class: 12

Time Allowed: 45 minutes

Maximum Marks: 10

All questions are compulsory unless stated otherwise.

Name: \_\_\_\_\_

Roll Number: \_\_\_\_\_

Class: 12 Section: \_\_\_\_\_

## Section A: Multiple Choice Questions

Attempt all questions. Each question carries 1 mark.

1. [Easy] Which equation correctly represents linear motion with constant acceleration? [1 Marks]
2. [Easy] The torque produced by a force  $F$  applied at a distance  $r$  making an angle  $\theta$  with the lever arm is given by? [1 Marks]
3. [Easy] In rotary motion, angular velocity  $\omega$  is related to linear velocity  $v$  by which formula? [1 Marks]
4. [Easy] The force required to produce a linear acceleration  $a$  on a mass  $m$  is given by? [1 Marks]

## Section B: Short Answer Questions

Answer each question. Each question carries 2 marks.

1. [Moderate] Derive the expression for the final velocity  $v_f$  of an object moving with initial velocity  $v_i$  under constant acceleration  $a$  after time  $t$ . Provide the formula and a brief explanation. [2 Marks]
2. [Moderate] Explain how torque is related to force and the perpendicular distance from the axis of rotation. Include the formula and describe the role of the sine of the angle between force and lever arm. [2 Marks]
3. [Moderate] A particle moves in a straight line with an initial velocity of 5 m/s and constant acceleration of 2 m/s<sup>2</sup>. Calculate the displacement after 3 seconds using the appropriate kinematic equation. [2 Marks]

End of Question Paper

## Answer Key:

1.  $s = v_i t + \frac{1}{2} a t^2$
2.  $\tau = r F \sin\theta$
3.  $v = \omega r$

4.  $F = m a$

5.  $v_f = v_i + a t$ . From  $a = (v_f - v_i)/t$ , rearranging gives  $v_f = v_i + a t$ .

6.  $\tau = r F \sin\theta$ , where  $r$  is the distance from the axis,  $F$  is the magnitude of the force, and  $\theta$  is the angle between  $r$  and  $F$ . The  $\sin\theta$  term ensures only the component of force perpendicular to the lever arm produces torque; maximum torque occurs at  $\theta = 90^\circ$ .

7.  $s = v_i t + (1/2) a t^2 = 5 \cdot 3 + 0.5 \cdot 2 \cdot 9 = 15 + 9 = 24 \text{ m}$ .